Blind, Pre-Registered Validation of LLM Population Simulation:
A Pass, Its Mechanism, and a Falsification

Umar Aslam*

Independent researcher

`https://github.com/Umaraslam66/Agora`

July 2026 — v1.0

Abstract

LLM-based simulations of human populations are spreading faster than the standards used to validate them. We report what we believe is the first *blind, pre-registered, adversarially controlled* validation of an LLM population-simulation method against held-out natural experiments, with every metric, bar, and decision rule frozen and published before any blind quantity was scored — and with both a positive and a negative verdict sealed, whatever they said. The method seeds one persona per real household-travel-diary record, executes daily choices with a calibrated fast (non-LLM) executor governed by an LLM-written behavioral card, and lets an LLM slow brain rewrite cards under announced policy shocks. In the primary arena (the Seattle SR 99 tunnel tolling of 2019), the placebo-corrected blind prediction covered the observed -28% volume response (central drop 0.275, 80% interval $[0.267, 0.286]$ as fractions of baseline volume) and beat both the official forecast (-45%) and a no-LLM calibrated comparator (central 0.310, non-covering). A same-day mechanism audit, sealed as an amendment before the second arena fired, showed $\geq 98.6\%$ of the predicted response flowed through the calibrated route-choice channel and $\approx 1\%$ through LLM card rewrites: the method won through *calibration transfer*, not runtime persona intelligence. The frozen method was then transferred unchanged to a second blind arena (the Stockholm congestion charge, cross-instrument and cross-population by design). The transfer verdict is an unambiguous, clean null: the card channel prices at $\sim 0.05\%$ of baseline against a real-world $\sim 21\%$ effect, the pre-registered hysteresis signature fails at every sensitivity threshold, and yoked placebo arms show the null is not drift. A non-blind information-value analysis in equivalent-sample-size units completes the picture at the individual level: persona cards carrying no behavioral evidence are worth fewer than ten real diary records at individual prediction, the value of an observed day is almost entirely replay rather than generalization, and a deterministic template card built from the same diary strictly outperforms the LLM-written card. We conclude that what was demonstrated is calibration transfer through a disciplined harness; the LLM adaptation channel, as built, carries no measurable price response — and that the pre-registration harness itself, which caught all of this, is the main contribution.

*Draft prepared with agent assistance under the project's standing rules: agents draft, the owner seals. All blind verdicts referenced here were sealed before this paper was written and cannot be altered by it.

1 Introduction

Generative-agent simulation is having a moment. Since interactive LLM-agent societies were first demonstrated (Park et al., 2023), and especially since LLM agents seeded from interview transcripts were shown to reproduce survey responses of a thousand real individuals (Park et al., 2024), a growing body of work treats LLM populations as instruments for social-science measurement, policy piloting, and synthetic survey research. The validation standard in this literature is, almost uniformly, *fit to held-out survey answers of the same kind the model was seeded on*, scored by researchers free to iterate until the fit looks right.

The strands of work this builds on are by now well established. Argyle et al. (2023) showed that a suitably conditioned language model reproduces the response distributions of demographically defined human subgroups ("silicon samples"). Aher et al. (2023) introduced Turing experiments, replicating classic economic and social-psychology studies with simulated participant samples. Horton et al. (2023) treat the LLM as a simulated economic agent — *Homo silicus* — to be endowed and experimented on in silico. And the agent-based-modeling community has debated empirical validation standards for exactly this class of simulation since well before LLMs entered it (Windrum et al., 2007).

That standard has three well-known failure modes. First, it cannot distinguish memorized correlation from behavioral mechanism: the events against which such simulations are validated are public and famous, and the models were trained on descriptions of them. Second, it is not blind: metric choice, threshold choice, and re-run decisions are made after seeing results. Third, it rarely carries a falsification arm: a cheap non-LLM baseline scored under identical protocol, so that the marginal value of the LLM can be isolated rather than presumed.

This paper reports a validation exercise built to fail honestly on all three axes, applied to a concrete, decision-relevant domain: urban travel behavior under road pricing. The design principles were:

1. **A frozen pre-registration outranks everything.** Every metric (E1–E7), bar, decision rule, correction, and mechanism choice was committed to a public repository before any blind quantity existed; changes are append-only, dated amendments (A1–A8), each sealed before the quantity it governs was scored, and *pushed* before firing — commit dates are forgeable, push dates are not.
2. **Blind tests fire once.** Each arena's blind battery was scored a single time against a quarantined truth series and sealed, whatever it said. Re-runs are post-hoc by definition and cannot overwrite a verdict.
3. **Adversarial controls.** A calibrated multinomial-logit falsification arm for ordinary behavior; a no-LLM comparator arm for the blind shock; a yoked, content-free placebo arm defining every scored response as a drift-corrected difference; masking with a versioned forbidden-token list and contamination probes for memorization; and a measured two-leg drift rule that can flag any verdict as drift-dominated.
4. **Negative results are product.** The pre-registration binds the project to publish failures with the same prominence as passes; deleting one is falsification.

Under this harness the method passed its first blind test — and the harness itself then dismantled the flattering interpretation of that pass. A sealed same-day audit showed the pass was carried by a calibrated aggregate dial, not by runtime persona intelligence (§4); the second

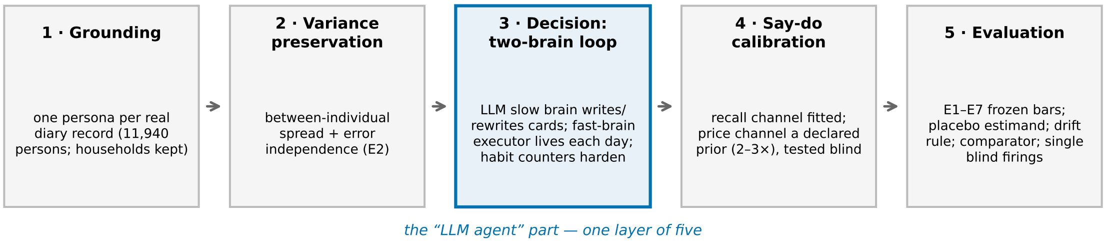

Figure 1: The five method layers. The two-brain loop (layer 3) is one layer of five; the blind-test mechanism audit (§4.3) shows the calibrated world model of layers 1–2 and the say-do correction of layer 4 carried the blind pass.

blind arena, constructed so that the persona-adaptation channel had to do the work alone, returned a clean null (§5); and individual-level scoring shows the LLM card adds no predictive value over a deterministic template built from the same evidence (§6). We regard the resulting record — one pass with its mechanism correctly attributed, one falsification, and the harness that forced both — as the contribution.

2 The method: five layers

The method is a pipeline of five layers; the LLM "two-brain" loop that gives such systems their name is deliberately only one of them (Figure 1).

Layer 1 — Grounding (1:1 record seeding). Every persona is seeded from exactly one real respondent of the Puget Sound Regional Council household travel survey (2017 and 2019 waves; 11,940 persons in 6,319 households), preserving the household structure. There is no sampling from fitted marginals: each simulated person *is* a masked compression of one real person's diary, demographics, and stated-attitude module.

Layer 2 — Variance preservation. The pipeline is scored (eval E2) on preserving between-individual variance and error independence — a hidden "single mind" fails E2(ii) even if means are right. Sealed result: spread ratios 0.987/1.002/0.998 on the three pinned dimensions (band $[0.8, 1.2]$), error correlation ≈ 0 (bar ≤ 0.20).

Layer 3 — Decision (the two-brain loop). A *slow brain* (Qwen3-8B, temperature 0) writes a compact persona card — day patterns, behavioral rules, habit counters — at initialization and rewrites it on surprises; a *fast brain* (plain code, no LLM) executes every daily choice from the card through a calibrated route/mode choice layer. Per-rule habit-strength counters harden with repetition; past a calibrated threshold a rule becomes immutable and a rewrite must restore it verbatim. Rewrites pass a five-gate validator (schema, mask-lint, replay-smell, feasibility, fidelity-to-own-diary); under an announced shock the fidelity gate is dropped (a gate scored against the pre-shock diary would clamp the adaptation being measured) and drift control moves to the placebo arm (below).

Layer 4 — Say-do calibration. Stated-vs-revealed discrepancies are handled as two channels sealed in advance: a *recall* channel fitted person-level on stated/revealed pairs inside the seeding data, and a *price* channel declared as a literature prior (revealed toll response 2–3× stated; Brownstone and Small, 2005), never fitted locally — the pre-registration explicitly forbids fitting it in-arena and gives it exactly one test, the blind E3(iii) transfer clause.

Layer 5 — Evaluation. Seven eval families (E1–E7) with frozen bars: grounding fidelity against a calibrated MNL falsification arm (E1), variance (E2), say-do (E3), blind shock prediction (E4), contamination probes (E5), hysteresis (E6), and an information-value ablation (E7). The sealed E1 verdict: pooled TVD 0.0614 (bar ≤ 0.10), beating the MNL arm (0.1125) with the paired 95% CI of the difference entirely below the sealed $\varepsilon = 0.00655$.

3 Study design: two blind arenas under one wall

Arenas. The primary arena is the Seattle SR 99 tunnel tolling event: tolling began 2019-11-09 on a tunnel with an excellent untolled bypass; the blind window ends 2020-02-29 (pandemic cutoff). Blind test 1 (BT1) scores the tolling onset; there is no Seattle second test. The transfer arena is the Stockholm congestion charge (2006 trial: on, off, permanently on), scored as blind test 2 (BT2) with the frozen method applied unchanged. The transfer is deliberately *cross-instrument* (link toll vs. cordon charge), *cross-country*, and *cross-population*: Stockholm personas are the PSRC-seeded personas reweighted to published Stockholm-County marginals (labeled METHOD-TRANSFER), a sealed design choice that makes BT2 a test of the method rather than of source-population idiosyncrasy — and the amendment sealing it (A1.3) explicitly forbids re-attributing a transfer failure to seeding class after the fact.

The wall. Nothing measured after 2019-11-08 (primary) or after the Stockholm P0 baseline (transfer) may influence any tuning, prompt, threshold, correction, or model choice. The truth series is import-quarantined (`evaluation/truth/`) from all agent-facing code, enforced by a static-scan plus runtime-tripwire CI test; the blind firing entrypoints are guarded by a pre-execution hook requiring the owner's inline authorization. Both firings were single SLURM jobs on EuroHPC Leonardo (BT1: 10h19m; BT2: 16h02m), fired on the owner's explicit go, with all governing amendments public before submission.

Masking. Agents live in "City K": masked zone codes, shifted dates, credit-denominated prices preserving per-period real ratios, a versioned forbidden-token list (v0.3, both arenas' real vocabulary) enforced in CI on every agent-visible byte. Contamination probes (E5) run continuously: the masked-vs-unmasked paired A/B showed a 4.6% unmasked advantage (flag threshold 25%), and the fictional-price probe showed strictly monotone price response with no reproduction of the famous historical aggregate at non-historical prices.

The placebo doctrine. Every scored response is $\Delta Q = Q_{\text{policy}} - Q_{\text{placebo}}$, paired across common-random-number ensemble members. The placebo arm yokes the *trigger* and nulls the *reason*: the same agents receive, on the same day, a content-free reconsideration notice with all price fields nulled, in a world whose costs never change — so a faithful machine has nothing to do, and anything it does anyway is measured machinery drift and subtracted. The identifying assumption (equal drift across arms) is stated in the pre-registration as unverifiable by design, with its known failure direction (overstating the response) pinned before firing. A measured *two-leg drift rule* guards every verdict: a phase is flagged drift-dominated if the placebo magnitude exceeds twice a pre-measured floor (anomaly leg) or reaches half the policy response (absolute leg). The floors were measured and published before each firing (primary: 0.0254 on the SR 520 rehearsal; transfer: 0.006330 on a placebo-only baseline rehearsal).

The comparator. A sealed no-LLM comparator arm (A5) replays each persona's observed diary verbatim and responds only through the same frozen route-choice equilibrium, with its own value-of-time scale calibrated by the same procedure on the same anchor. Its prediction was frozen before firing: central 0.3096, 80% interval $[0.2980, 0.3212]$. It is reported, never a pass bar — with the interpretation of either outcome pinned in advance so neither side could be reframed post hoc.

4 Blind test 1: the pass, and what actually carried it

BT1 fired once (submitted 2026-07-18 on the owner's explicit authorization; run to completion and sealed 2026-07-19; Leonardo job 49744291) and scored, together: E4 on the T5 headline population, all five E7 tiers plus the T4-noclaims arm, the placebo arms, the tail-off bound, the comparator, and the E6 threshold band. Frozen inputs verified at load: habit threshold $\theta = 22$, VoT scale 2.2238, say-do price correction 2.5×, drift floor 0.0254.

4.1 The sealed E4 verdict

Sealed bar (on ΔQ, A4.2)	Value	Verdict
Observed −28% inside 80% interval	$0.28 \in [0.2673, 0.2864]$	**PASS**
Central closer than the −45% forecast	0.0048 vs 0.1700	**PASS**

Table 1: BT1 headline (T5, sole headline per A4.1): ΔQ central 0.2752, 80% interval $[0.2673, 0.2864]$ against the observed −28% tunnel-volume drop. The two-week trajectory point (0.2798 $[0.2651, 0.2979]$ vs. observed −26%) also covered.

The comparator missed: its interval $[0.2980, 0.3212]$ does not cover 0.28, and its central is 6× farther from the observation than the method's (0.0296 vs 0.0048). The E3(iii) transfer clause — the frozen 2.5× price correction must improve the blind central versus the uncorrected ablation — passed by 0.0004 of drop: recorded plainly as surviving its test with near-nil measured added value in this arena. Drift verdicts: the headline arm's placebo/toll ratio was 0.006, four times *below* the rehearsal floor; of six populations, only T3 tripped both drift legs (§4.2).

4.2 The E7 tier curve on the blind shock

E7 rebuilt the full population at five nested information tiers — T1 demographics only, T2 +world context, T3 +stated claims, T4 +one observed diary day, T5 +the full trace — CRN-paired, identical generation pipeline, plus a T4-noclaims side arm (behavior without stated claims). All tiers fired inside the single BT1 job.

Two anomalies stand out: the curve is *flat* (a demographics-only tier covers the blind truth), and T4-noclaims sits *above* T4 (withholding stated claims moved the central +0.85pp). Both were attacked before any interpretation was allowed to stand.

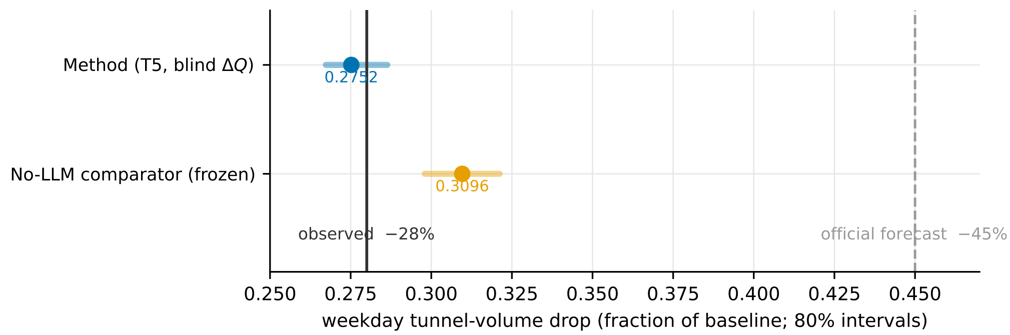

Figure 2: BT1, sealed: the method's blind 80% interval covers the observed −28%; the frozen no-LLM comparator misses high; the official forecast (−45%) is far low. All quantities are placebo-corrected ΔQ per the sealed estimand.

Tier	ΔQ central	80%	covers 0.28	drift-dominated
T1	0.2748	[0.2673, 0.2856]	yes	no
T2	0.2741	[0.2645, 0.2843]	yes	no
T3	0.2078	[0.1998, 0.2171]	**no**	**YES**
T4	0.2798	[0.2693, 0.2921]	yes	no
T4-noclaims	0.2883	[0.2772, 0.3001]	yes	no
T5	0.2752	[0.2673, 0.2864]	yes	no

Table 2: E7 on BT1 (diagnostic; T5 is the sole headline). The curve is flat except T3 — and §4.3 explains both facts.

4.3 The sealed mechanism audit: dial-dominated

A same-day, owner-run audit (sealed as amendment A7 before the second arena fired) decomposed every arm's response exactly into a *demand term* (corridor car-travel leaves — the only channel card rewrites can move) and a *route term* (share shift among remaining drivers under the shared frozen VoT dial), replaying sealed arms deterministically (drop reproduction $< 10^{-9}$; per-agent facility reconstruction exact in 8/8 arms).

In every covering tier at least 98.6% of the scored response flows through the fast-brain route choice under the shared calibrated dial — a channel that is tier-invariant by construction. The card-rewrite channel, the only channel tier information can touch, carried ≈ 1%. The tiers genuinely differ (0% identical cards across tiers; added-rule similarity to T5 of only 0.02–0.35), so the flat curve is not accidental sameness: different personas produce different adaptations, feeding a channel that carries one percent of the response. The flat curve is arithmetic, not a discovery about information value.

The comparative reading against the no-LLM comparator was corrected accordingly, and the sealed amendment requires every future summary to carry it verbatim: *"the two-brain method beat the aggregate comparator through calibration transfer — the adaptation channel's presence at fit time moved the dial to the right place — not through runtime persona intelligence."* Concretely: the method's VoT dial was jointly calibrated with the LLM adaptation channel present and the say-do correction applied, landing at scale 2.2238, which transfers to 0.275 at the blind rates; the comparator's solo calibration landed at 1.8512 → 0.310, outside its own interval. BT1 may not be presented as evidence that persona-level simulation adds predictive value over a

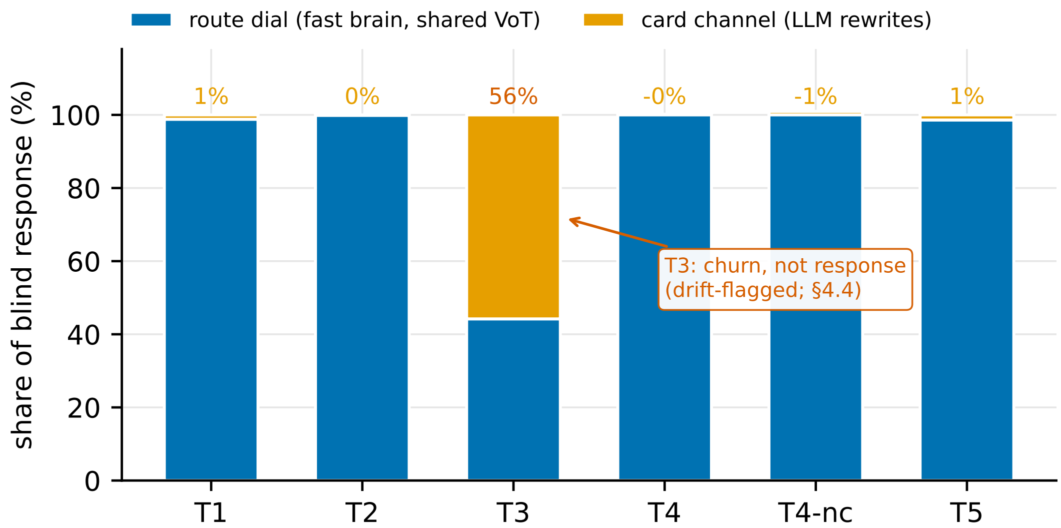

Figure 3: The sealed two-term decomposition of the BT1 response (audit A7): in every covering tier $\geq$ 98.6% of ΔQ flows through the fast-brain route-choice dial; the card-rewrite channel carries $\approx$ 1%. T3's anomalous split is churn, not response (§4.4).

well-calibrated aggregate model in this arena.

4.4 The T3 instability: stated claims without behavior

T3 — stated claims with no observed behavior — is the sole non-covering tier and the sole drift-flagged tier, and the audit showed these are one phenomenon. Rewriting a claims-only card regresses it toward the claims *regardless of notice content*: the toll and placebo arms produced identical card-driven outcome-class splits, and the placebo arm alone lost 25% of its corridor car-driver base (535 $\rightarrow$ 400 drivers/day) under a notice containing nothing. CRN pairing cancels the churn in ΔQ almost exactly, but the churn shrinks the toll arm's driver base by $\sim$ 25%, deflating the paired route response proportionally ($0.275 \times 0.75 \approx 0.207 \approx$ the sealed 0.2078). The sealed two-leg drift rule flagged exactly this tier unaided — the control doing precisely its job. Claims-without-behavior cards are unstable under *any* rewrite trigger; we return to this at the individual level in §6.

5 Blind test 2: the falsification

BT2 is the arena the adaptation channel could not hide in. The Stockholm cordon instrument has *no route dial by construction*: crossing the cordon is determined by origin and destination; there is no priced route with an unpriced twin. Whatever response exists must run through the card channel — mode change, suppression, re-timing. The transfer inventory pre-declared exactly this reading, and amendment A8 pinned the four-phase protocol (P0 baseline, P1 introduction, P2 removal — the hysteresis phase, P3 permanent return), the yoked placebo, the estimand, a pre-measured arena drift floor (0.006330, published standalone before firing), and the requirement that the channel decomposition ship with the verdict.

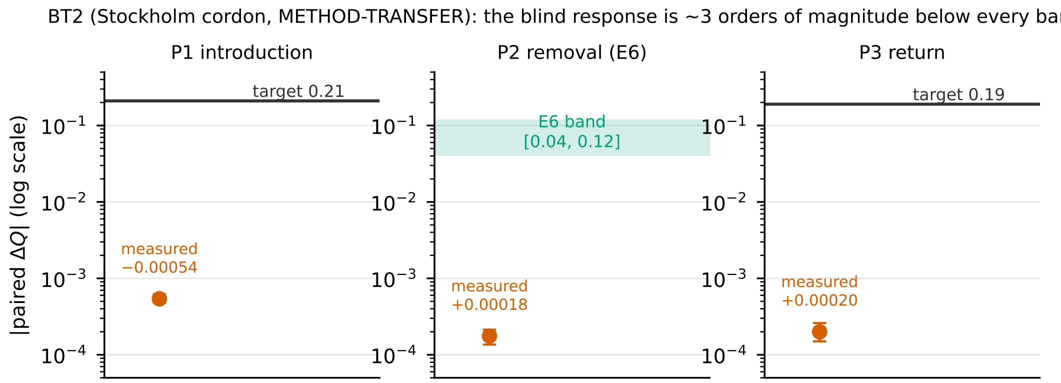

Figure 4: BT2, sealed: measured paired ΔQ per phase (note the magnified scale) against the pre-registered targets and the E6 band. The response is $\sim$ 200–400× smaller than required; placebo magnitudes sit at the pre-measured floor, so the null is not drift.

BT2 fired once (2026-07-20, Leonardo job 49860805, 16h02m) on the owner's explicit inline authorization, after the pre-firing declaration was pushed. The sealed verdict:

Quantity (sealed bar)	Bar	Measured (paired ΔQ, 80%)	Verdict
P1 introduction vs truth 0.21	covered	-0.00054 $[-0.00062, -0.00048]$	**FAIL**
P3 return vs truth 0.19	covered	$+0.00020$ $[+0.00015, +0.00026]$	**FAIL**
E6 arm (a) P2 residual	$\in [0.04, 0.12]$	$+0.000176$ $[0.000136, 0.000213]$	**FAIL** ($\sim$200× low)
E6 arm (b) memory-ablated	< 0.04	$+0.000176$ (identical to $\sim 10^{-13}$)	vacuous
E6 non-overlap	intervals disjoint	identical	**FAIL**
E6 threshold band $\{6..26\}$	per threshold	ΔQ identical at every θ	**FAIL at all**
Drift P1/P2/P3	$\leq 2 \times 0.006330$, < 0.5	0.00647/0.00651/0.00599	**CLEAN**

Table 3: BT2, sealed (all quantities METHOD-TRANSFER-labeled): the method's blind response in the cordon arena is null — three orders of magnitude below every bar — and the null is clean, with the drift control green in all phases.

What the null is made of. Every crossing agent received and processed the announcements (3,999 accepted rewrites per arm-run, charged and placebo alike). The memory-ablated arm's unconstrained rewrites churned behavior substantially ($\sim$ 3.5% raw drops) — but *identically* under the priced and the nulled notice, so pairing cancelled it: churn, not response. E6's pre-registered contingency reading applies verbatim: arm (a) $\approx$ arm (b) $\Rightarrow$ "memory not load-bearing" — there is no hysteresis because there is no response for habit memory to preserve. The A8.4 decomposition confirmed the pre-declared *form* (share terms equal across arms within CRN noise) while the demand terms carry no separation because none exists. The card channel, isolated at last, prices at $\sim$ 0.05% of baseline against the real-world $\sim$ 21%.

Reading the two verdicts together. BT1's mechanism audit said the response ran through the dial; BT2, an arena with no dial, measured what the cards do alone: nothing. The A7 mechanism reading is thereby blind-confirmed from the opposite side. A method whose BT1 pass had genuinely run through persona adaptation would have moved here; it did not. The falsification is clean, pre-registered, and stands sealed beside the pass with equal prominence —

which is exactly what the harness was built to guarantee.

6 Information value at the individual level: the E7 ordinary-day ESS analysis

The blind results measure aggregates. The remaining pre-registered E7 analysis (A4.1; non-blind, scored after both arenas sealed) asks the individual-level question: *how much does each increment of persona evidence help predict a real person's own diary days* — scored in equivalent-sample-size (ESS) units (Gao et al., 2026): tier k's cards are worth ESS(k) real diary records if a flexible baseline trained on that many records matches their per-person predictive loss. The per-persona loss adapter, the baseline (the E1 MNL arm's day-structure and mode-choice components, refit per training block), the candidate grid, and both scoring targets were pinned in the E7 manifest before any ESS quantity was computed. The per-persona loss is the mean TVD, over the three frozen E1 families, between the arm's ensemble-predicted distributions and the persona's own observed days — on two targets: *all days* (identical across tiers; for behavior-carrying tiers partly in evidence, i.e. compression), and *held-out days* (multi-day personas only, excluding the pinned T4-shown day; a true generalization test for T4).

Arm	loss (all days)	ESS (all days)	loss (held-out)	ESS (held-out)
T1 (demographics)	0.660	≤ 10	0.645	≤ 10
T2 (+context)	0.659	≤ 10	0.644	≤ 10
T3 (+stated claims)	0.692	≤ 10	0.673	≤ 10
T4 (+one day)	0.146	> 5000	0.510	$\lesssim 10$
T4-noclaims	0.147	> 5000	0.506	≈ 20 (≥ 11)
T4-nofidelity	0.215	> 5000	0.516	≤ 10
T5 (full trace)	0.097	> 5000	$0.221^\dagger$	$> 1000^\dagger$
Deployed (M2) cards	0.114	> 5000	$0.216^\dagger$	$> 1000^\dagger$
Template cards	0.047	> 5000	$0.193^\dagger$	$> 1000^\dagger$

Table 4: E7 ordinary-day scoring at the individual level (11,940 personas; held-out target on the 2,975 multi-day personas). Loss = mean per-persona TVD (lower is better); ESS = equivalent real diary records (plugin / one-sided 95% lower bound). †held-out days in evidence for these arms (flagged per the pinned adapter).

The curve answers the individual-level question in three sharp statements. **First: without the person's own behavior, cards are worth less than ten strangers' records.** T1–T3 sit at losses of 0.66–0.69 against a 10-record baseline at 0.59; the pinned sequential test hands all three an ESS of ≤ 10 on both targets — and stated claims make the card *worse* than demographics alone (T3 0.692 vs T1 0.660), the individual-level face of the T3 instability the blind test flagged (§4.4). **Second: the value of behavioral evidence is almost entirely compression, not generalization.** On the all-days target the behavior-carrying arms are worth more than the entire grid ($>$ 5,000 records — unsurprising and flagged: their evidence contains the scored days, and the baseline's learning curve flattens near 0.53, so no training size reaches them). But on genuinely held-out days the one-shown-day card collapses from 0.146 to 0.510, worth $\lesssim$ 10 records — statistically indistinguishable from a baseline trained on ten strangers — and withholding the stated-claims module *improves* generalization (T4-noclaims $\approx$ 20 records, the only tier that resolves above the grid floor). Knowing one of a person's days

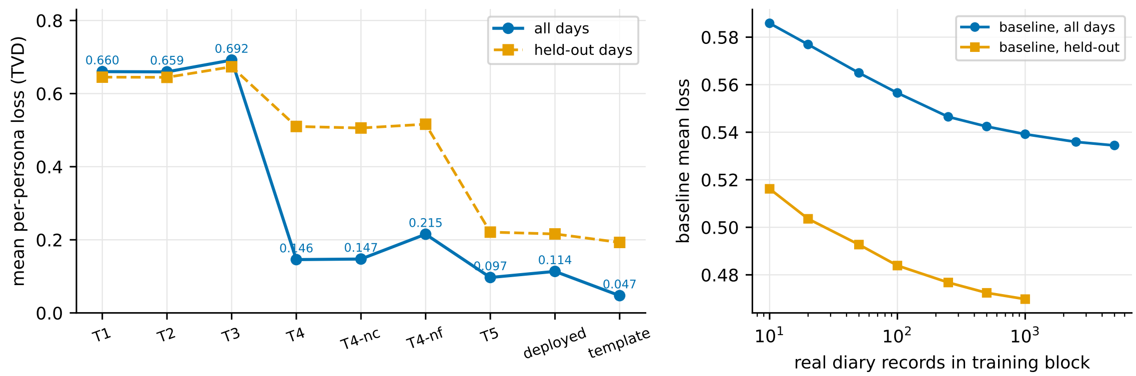

Figure 5: Individual-level information value. Left: mean per-persona loss by arm on both targets. Right: the flexible baseline's learning curve — it flattens near 0.53 (all days) / 0.47 (held-out), far above every behavior-carrying arm: no amount of *other people's* records substitutes for a person's own observed behavior, which is why the behavior-carrying arms exceed the ESS grid on the compression target.

buys near-nothing about their other days beyond what population statistics already provide. **Third: the fidelity gate, not the LLM, carries a large share of the behavioral tiers' value.** Switching the generation-time fidelity gate off (T4-nofidelity) degrades the loss from 0.146 to 0.215 and the aggregate TVD from 0.047 to 0.115 — consistent with the sealed M2 measurement of the gate's worth ($\approx$ 0.07 TVD), and further evidence that the verification machinery around the model, not the model's authorship, does the work.

Two cross-checks anchor the analysis to the sealed record: the deployed-population aggregate diagnostic reproduces the sealed E1 verdict to 10^{-4} (pooled TVD 0.0613 here vs the sealed 0.0614), and the tier populations are byte-identical (SHA-256) to those fired in BT1. The aggregate diagnostic also yields the starkest dial-domination exhibit in the project: **T1's ordinary-day aggregate TVD is** 0.68 **— seven times the E1 bar; as a deployed population it would fail E1 outright — yet its blind ΔQ covered the observed** -28% (Table 2), because the paired, CRN-matched estimand rides entirely on the shared calibrated dial. Coverage of the blind aggregate was never evidence of persona realism.

Template vs. LLM head-to-head. The deployed population contains 8,496 LLM-written cards and 3,444 deterministic template (fallback) cards; a full template population exists for every persona. Paired per-persona on the 8,496 LLM-card personas, the LLM card is *worse* than the template card by 0.095 mean loss (household-atomic bootstrap 95% CI $[0.092, 0.098]$) on the all-days target and remains worse by 0.054 $[0.046, 0.062]$ on the held-out target. This mirrors, at the individual level, the sealed M3 aggregate diagnostic (template pooled TVD 0.043 vs LLM 0.074) — and completes the arc of §4.3 and §5: at no level of analysis did the LLM's authorship of the behavioral card demonstrate measurable predictive value beyond the evidence it was handed.

7 Two agents, up close

Boxed case study: one honest adapter, one churner (both real, from the sealed BT1 inspector export; masked vocabulary verbatim)

P05036 (T5, full information). A suburban two-car household's main driver; every observed trip by car. Its card: two patterns (commute/errand day) and two rules — "work → car", "home → car". Both rules crossed the immutability threshold weeks before the shock. On announcement day the slow brain rewrote the card; the entire diff re-timed the work departure from night to the morning peak; the locked rules came back verbatim. Over 63 charged days the policy twin and the placebo twin differed on *four* days — all route choices, priced crossing vs. parallel road. Multiply that handful of nudged days across thousands of drivers and you get the $\approx 1\%$ card channel of the sealed audit — visible, sensible, small.

P00169 (T3, claims only). The same kind of record, but its card was written from stated claims alone ("I usually commute by transit"), no diary. On announcement day *both* arms' rewrites simplified the two transit rules by dropping their "work" qualifier — turning "I take transit to work" into "I take transit at peak, whatever I'm doing." With no behavioral anchor, the gate had nothing to hold the card against. The twins' 102-day timelines are byte-identical: the "response" to the charge equals the response to a notice containing nothing. Hundreds of T3 agents did the same, in both arms — which is why T3, alone, was flagged by its own placebo.

One agent with full information adapts believably and barely; one with only self-description "adapts" identically no matter what it is told — noise wearing the costume of a response, caught by the control. This is the information-value result in miniature: behavioral detail earns its keep as an *anchor against churn*; the method's measured intelligence lives in its calibrated daily choices, not in dramatic persona rewrites.

8 Limitations

Sealed caveats are restated here as the records require; none is implicated in either verdict's direction.

1. **Two arenas.** $n = 2$ blind batteries, one instrument each. The corridor arena is nearly a pure route-choice problem (the $\approx 1\%$ card channel is a statement about that estimand, not about persona loops generally); the cordon arena isolates the card channel completely. Other estimands (long-run equilibria, land use, non-price policies) are untested.

2. **method-transfer seeding.** Stockholm personas are PSRC-seeded and reweighted to published marginals — the sealed, deliberately harder claim. The child-share rider is restated verbatim: *"children are unconstrained by M1 and follow their households; weighted child share 13.7% vs arena 21.9% (2005). Expected direction of bias: mode-share composition (escort travel, within-household car allocation in the baseline crossing mix),* ***not*** *the price response — the priced adaptation channel runs through adult decision-makers under the frozen constants, which reweighting never touches."*

3. **Truth-side caveats.** The Stockholm P2 anchor carries the published roadwork confound (*"off-year residual partially confounded by roadwork at two bridges"*), which is why the E6 band was widened ±2pp at sealing; the BT1 -28% carries an unquantified seasonal component the placebo cannot correct (pinned pre-firing). The measured BT2 miss ($\sim 200\times$ below band) dwarfs both.

4. **Population-margin approximations, sealed with the build:** the M2 central-cell proxy (40.8% $\to$ rings {core, inner}); M5 income excluded from raking (rank-preserving no-op without an invented currency mapping); the masked daily charge cap recorded, not enforced (binds only at ≥ 4 peak crossings/day).
5. **Say-do price prior untested by transfer.** E3(iii) passed in BT1 by 0.0004 — near-nil measured value — and its transfer reading was *mooted* by the BT2 null (with the card channel inert, the correction had no channel to act on): recorded as mooted, not answered.
6. **ESS caveats.** The ordinary-day ESS analysis is non-blind (scored after both arenas sealed, protocol pinned before scoring); for behavior-carrying tiers the all-days target is partly in evidence (compression, not generalization — the held-out target exists for exactly this reason); the household-atomic block adaptation to the pinned estimator is reported with realized block sizes.
7. **One model.** The slow brain is a single 8B open-weights model at temperature 0. Larger or different models might write cards that price-respond; the harness is indifferent — any such claim would require a new pre-registered arena, and the sealed verdicts here cannot be revised by one.

9 Conclusions

Three findings, in the order the evidence forces them:

1. **What was demonstrated is calibration transfer.** A grounded, variance-preserving population with a jointly calibrated choice dial predicted a blind natural experiment correctly and beat both the official forecast and a no-LLM comparator — because the adaptation channel's presence *at fit time* moved the dial to the right place. That is a real, useful, replicable result about building forecasting systems — and it is not persona intelligence.
2. **The adaptation channel, as built, was falsified.** Given an arena where only the cards could respond, they did not (0.05% of baseline against a 21% real effect), at every habit threshold, with clean controls. LLM card-rewriting under announced price shocks, in this implementation, does not carry a price response — and stated-claims-only personas are actively unstable, churning identically under signal and placebo.
3. **The harness is the contribution.** Every load-bearing correction in this paper — the mechanism reattribution, the T3 flag, the falsification, the mooted prior — was forced by machinery committed before results existed: frozen bars, single firings, yoked placebos, measured drift floors, adversarial comparators, quarantined truth, and negative-results-as-product. None of it would have surfaced under fit-to-held-out-surveys validation. We commend the harness, not the method, to the field: the interesting question is not whether LLM populations can fit data they have seen, but what survives when they are made to predict, once, blind, against controls built to catch exactly the failure that occurred.

Reproducibility

The pre-registration (with all eight amendments), the guard hooks, the masking gates, the eval harness, both sealed verdict records, the mechanism audit, and every number in this paper are in the public project repository, `https://github.com/Umaraslam66/Agora`; blind quantities are reproducible from the sealed `runs/` manifests (results SHA-256 pinned at firing).

Data availability

The seeding microdata are the Puget Sound Regional Council household travel survey (2017 and 2019 waves), openly downloadable from PSRC. The truth series — SR 99 tunnel volumes and the official forecast, and the Stockholm phase effects — are taken from the cited published sources (WSDOT, 2018–2021; Eliasson et al., 2009; Börjesson et al., 2012; Eliasson, 2014). Record-derived persona cards and household-level microdata are excluded from the repository by the project's privacy discipline (masking gates and the no-microdata rule).

References

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., and Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *Proc. UIST 2023*.

Park, J. S., Zou, C. Q., Shaw, A., Hill, B. M., Cai, C. J., Morris, M. R., Willer, R., Liang, P., and Bernstein, M. S. (2024). Generative agent simulations of 1,000 people. *arXiv:2411.10109v1*. (Retitled "LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals" in later revisions.)

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., and Wingate, D. (2023). Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3), 337–351.

Aher, G. V., Arriaga, R. I., and Kalai, A. T. (2023). Using large language models to simulate multiple humans and replicate human subject studies. *Proc. ICML 2023* (PMLR 202).

Horton, J. J., Filippas, A., and Manning, B. S. (2023). Large language models as simulated economic agents: What can we learn from *Homo silicus*? *NBER Working Paper* 31122 (rev. February 2026); arXiv:2301.07543.

Windrum, P., Fagiolo, G., and Moneta, A. (2007). Empirical validation of agent-based models: Alternatives and prospects. *Journal of Artificial Societies and Social Simulation*, 10(2), 8.

Gao, W., Han, S., and Liang, A. (2026). How well do LLMs predict human behavior? A measure of their pretrained knowledge. *arXiv:2601.12343*. (Equivalent-sample-size estimator pinned by amendment A4.1; block-out CV and sequential inference used exactly as published.)

Brownstone, D., and Small, K. A. (2005). Valuing time and reliability: assessing the evidence from road pricing demonstrations. *Transportation Research Part A*, 39(4), 279–293.

Eliasson, J., Hultkrantz, L., Nerhagen, L., and Smidfelt Rosqvist, L. (2009). The Stockholm congestion-charging trial 2006: Overview of effects. *Transportation Research Part A*, 43(3), 240–250.

Börjesson, M., Eliasson, J., Hugosson, M. B., and Brundell-Freij, K. (2012). The Stockholm congestion charges — 5 years on. Effects, acceptability and lessons learnt. *Transport Policy*, 20, 1–12.

Eliasson, J. (2014). The Stockholm congestion charges: an overview. *CTS Working Paper* 2014:7.

Washington State Department of Transportation / Washington State Transportation Commission. SR 99 tunnel tolling: pre-tolling forecast, rate adoption (2018-10-16), and one-year tolling report; SR 520 toll traffic and revenue records. Institutional sources pinned, with the reading most favorable to the forecast, in the public pre-registration (§7, A1.1/A3.3).